2 **(a)** The function $f(x)$ is defined by

$f(x) = \sqrt{1+2x}$ for $x \geqslant -\frac{1}{2}$.

Find an expression for $f^{-1}(x)$ and state the domain of this inverse function. **[3]**

(b) Explain why $g(x) = 1+x^2$, with domain all real numbers, has no inverse function. **[1]**